

Jiarong Gu

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Current Address: 7 Francis Darwin Court, Cambridge, UK

Relocating in September to: Schnorr-von-Carolsfeld-Strasse 5, 81927, Munich, Germany

EDUCATION AND EXPERIENCE

[2023-2025] King's College London [MSc Complex Systems Modelling] 2:1 (Pass with Merit)

- [MSc dissertation] The active voter model on a triangular grid
 - Simulated a 3-state active voter model on a triangular lattice.
 - Learnt to code in C for more direct management of data and memory.
- [Pattern Recognition and Deep Learning]
 - Experience with KERAS.
 - Created neural network in Python to classify MNIST data.
- [Theory of Complex Networks]
 - Diffusion processes on graphs and the Laplacian matrix.
 - Random graphs and the Erdős-Rényi model.

[2020-2023] The University of Manchester [BSc (Hons) Physics] 2:1 (Upper-second)

- [BSc dissertation] Dissertation on Evolutionary Game Theory (EGT) and Cancer.
 - Reviewed literature and reproduced results.
 - Simulated predator-prey dynamics in MATLAB.
- [Introduction to Nonlinear Dynamics]
 - Bifurcation diagrams, phase diagrams, non-dimensionalisation of ODEs.
 - Linear stability analysis of stationary points.
 - Population dynamics in discrete time and emergence of chaos.
- [Laboratory work]
 - Data collection and analysis, maintaining an organised lab space.
 - Calibrating sensitive instruments and handling radioactive material.
 - Collaboration and good communication with fellow students.
 - Notebook-keeping and report writing.
- [C++ coursework] Recreated an animated version of Conway's Game of Life.
 - Object-Oriented programming.
 - Code cleanliness and function organisation.
 - Time-management under tight deadlines.

[2017-2020] High School Affiliated to Renmin University of China

- A-Levels: A* in Physics
A* in Biology
A in Further Mathematics

POSITIONS OF RESPONSIBILITY

[Spring 2024, 2025 and 2026] **Kavli Institute of Cosmology, Cambridge**
[Open day demonstrator]

- Helped set up exhibits and activities, prepared materials for workshop.
- Ensured equal opportunities for every participant and mediated conflicts.
- Explained scientific concepts to wide range of audiences.

[Nov. 2023 – Jun. 2024] **King's College London**
[Student representative]

- Acted as liaison between student and faculty staff in a professional setting.
- Compiled and relayed messages in a timely manner.
- Devised solutions for issues or conflict.

[Oct. 2023 – May. 2024] **King's College London**
[Exhibition co-curator and invigilator]

- Presented the basics of Chaos Theory to fellow students and the commissioned artist.
- Compiled a video tutorial on the abacus using script and animation provided by exhibition lead.
- Voiced the script, spliced the animation and audio together in under a day.

[Sep. 2021- Sept. 2022] **The University of Manchester**
[Participant in atmospheric muon detector project]

- Built detector using organic scintillator.
- Soldered the detection circuit.
- Visualised detection rate against altitude.

PROGRAMMING LANGUAGES

[C] Self-taught during master's project for more direct management of data and memory.

[C++] Recreated Conway's Game of Life as coursework project.

[Python] Proficient in performing curve-fitting, training neural networks and analysing large data sets.

[MATLAB] Self-taught. Proficient in producing figures and handling large data sets.

[LaTeX] Self-taught. Proficient in type-setting and producing academic style reports.

OTHER

Created illustrations used in:

- University of Cambridge Research News: <https://www.cam.ac.uk/research/news/massive-black-hole-in-the-early-universe-spotted-taking-a-nap-after-overeating>
- Scientific American: <https://www.scientificamerican.com/article/gigantic-little-red-dot-threatens-to-upend-cosmic-history/>

Hobbies include hiking, bird watching, fencing, illustration and piano.